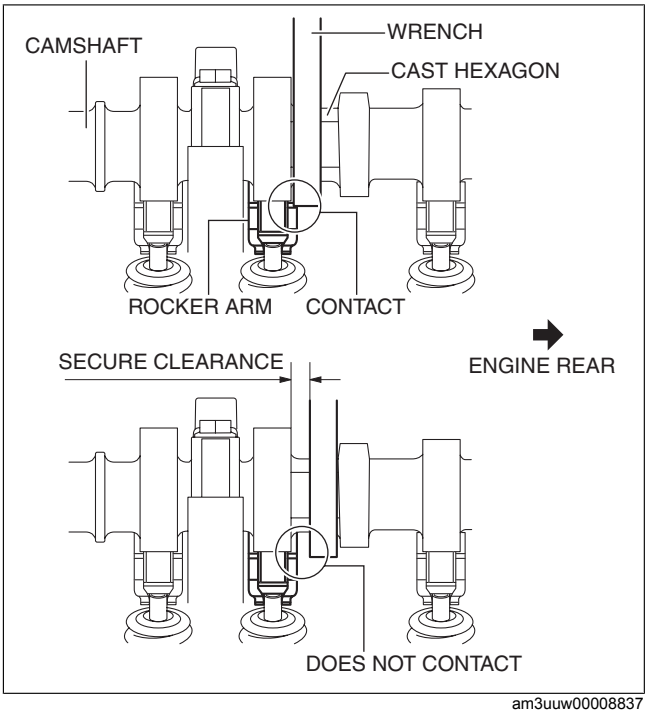
TIMING CHAIN DISASSEMBLY

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Caution

- **When rotating the camshaft using a wrench on the cast hexagon, the wrench may contact the rocker arm and damage the rocker arm. To prevent damage to the rocker arm when holding the camshaft on the cast hexagon, use the wrench at engine rear side as shown in the figure to secure a clearance between the cam.**

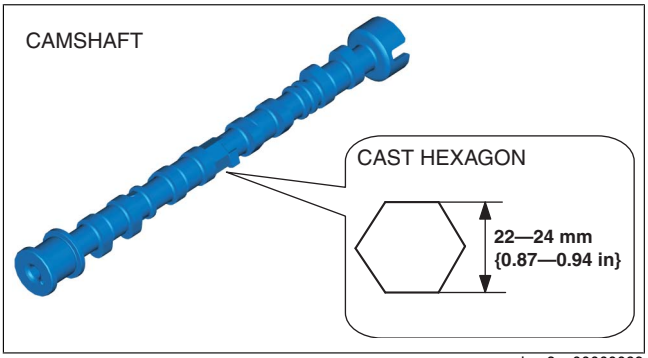


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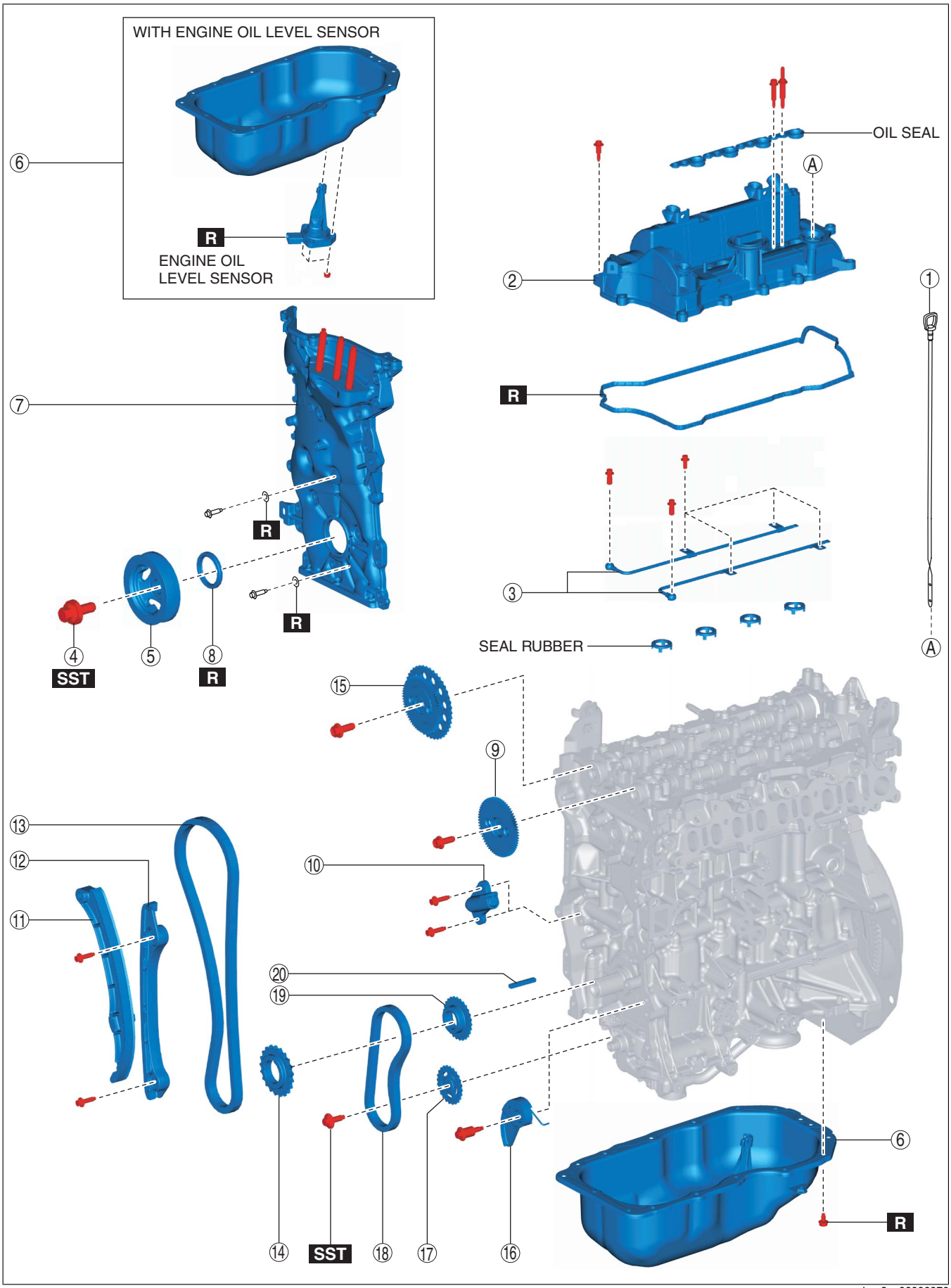
Note

- Width at the cast hexagon of the camshaft is 22—24 mm {0.87—0.94 in}.

1. Disassemble in the order indicated in the table.



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1	Dipstick
2	Cylinder head cover
3	Oil shower pipe

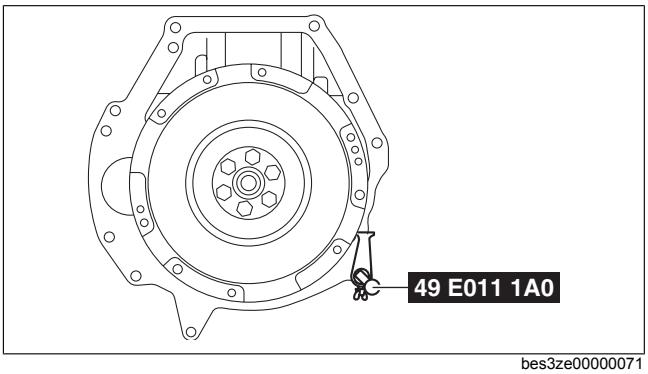
4	Crankshaft pulley lock bolt (See Crankshaft Pulley Lock Bolt Disassembly Note.)
5	Crankshaft pulley
6	Oil pan component (See Oil Pan Component Disassembly Note.)
7	Engine front cover (See Engine Front Cover Disassembly Note.)
8	Front oil seal (See Front Oil Seal Disassembly Note.)
9	Intake camshaft sprocket (See Intake Camshaft Sprocket Disassembly Note.)
10	Timing chain tensioner (See Timing Chain Disassembly Note.)
11	Timing chain tensioner arm (See Timing Chain Disassembly Note.)

12	Timing chain guide (See Timing Chain Disassembly Note.)
13	Timing chain (See Timing Chain Disassembly Note.)
14	Crankshaft sprocket (See Timing Chain Disassembly Note.)
15	Exhaust camshaft sprocket (See Exhaust Camshaft Sprocket Disassembly Note.)
16	Oil pump chain tensioner
17	Oil pump driven sprocket (See Oil Pump Driven sprocket Disassembly Note.)
18	Oil pump chain
19	Oil pump drive sprocket
20	Key

Crankshaft Pulley Lock Bolt Disassembly Note

1. Hold the crankshaft using the **SST**.

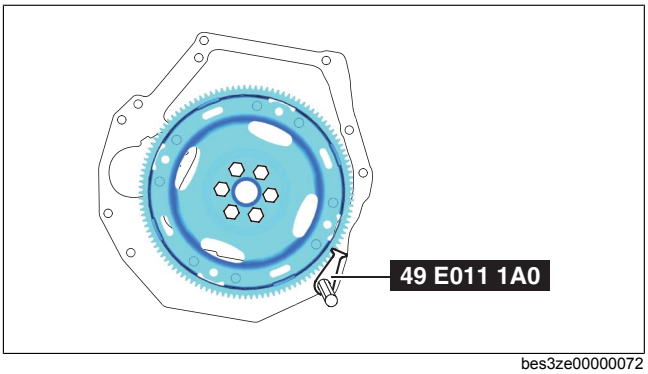
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2. Remove the crankshaft pulley lock bolt.



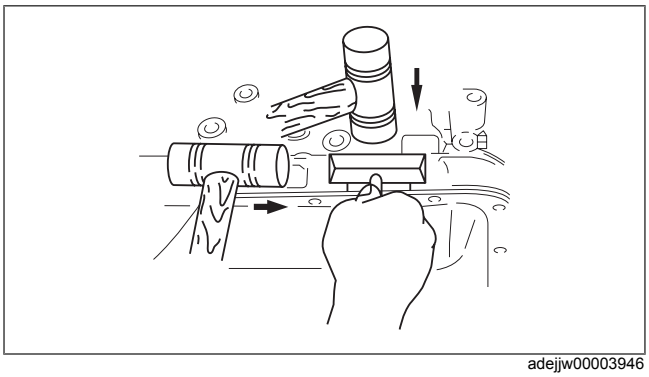
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Oil Pan Component Disassembly Note

Caution

- **If the engine oil level sensor is removed, it is necessary to replace it with a new one. (with engine oil level sensor)**

1. Remove the oil pan component using a separator tool.



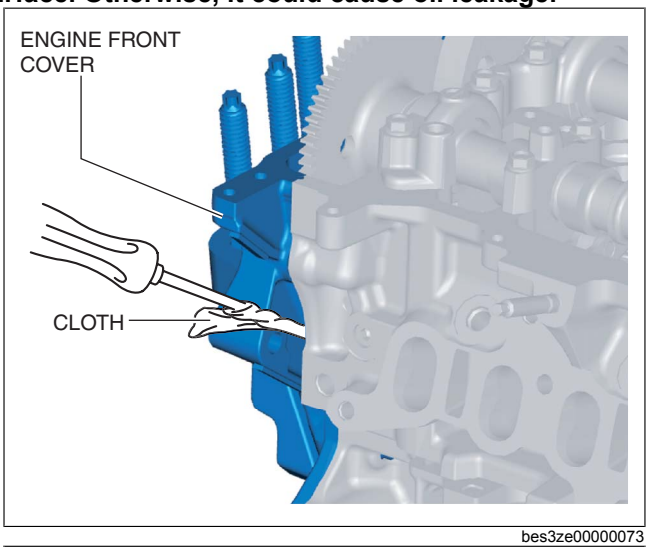
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Engine Front Cover Disassembly Note

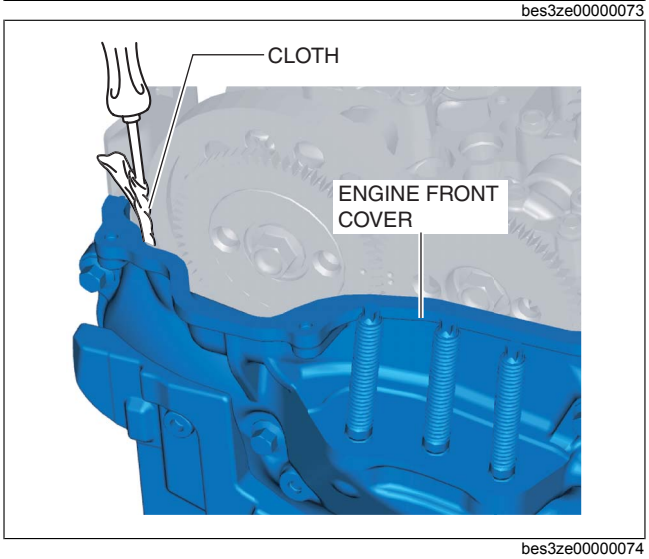
1. Remove the engine front cover installation bolts.
2. Using a screwdriver wrapped in a cloth, peel the sealant away a little at a time, and remove the engine front cover.

Caution

- **Do not apply excessive force to the screwdriver. Otherwise, the engine front cover could be damaged.**
- **Be careful not to scratch or damage the seal surface. Otherwise, it could cause oil leakage.**



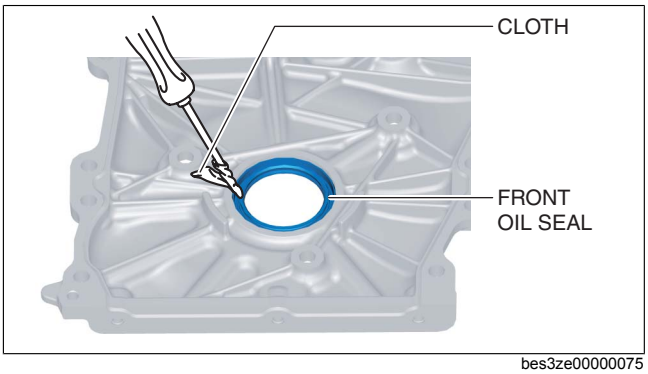
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Front Oil Seal Disassembly Note

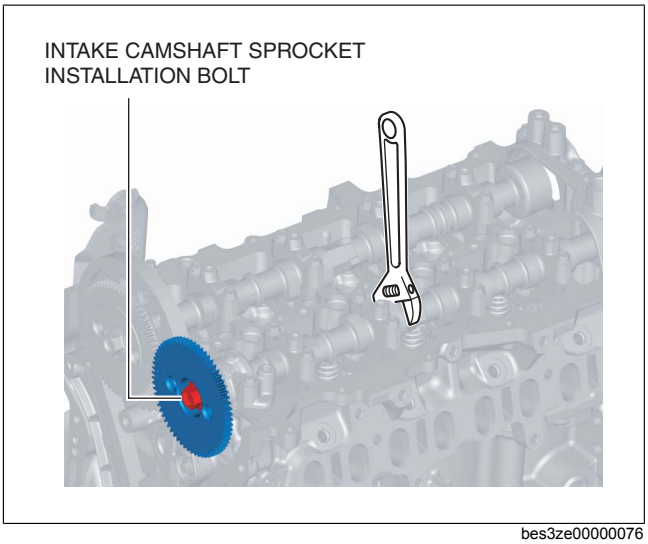
1. Remove the oil seal using a flathead screwdriver with the tip protected by a clean cloth.



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Intake Camshaft Sprocket Disassembly Note

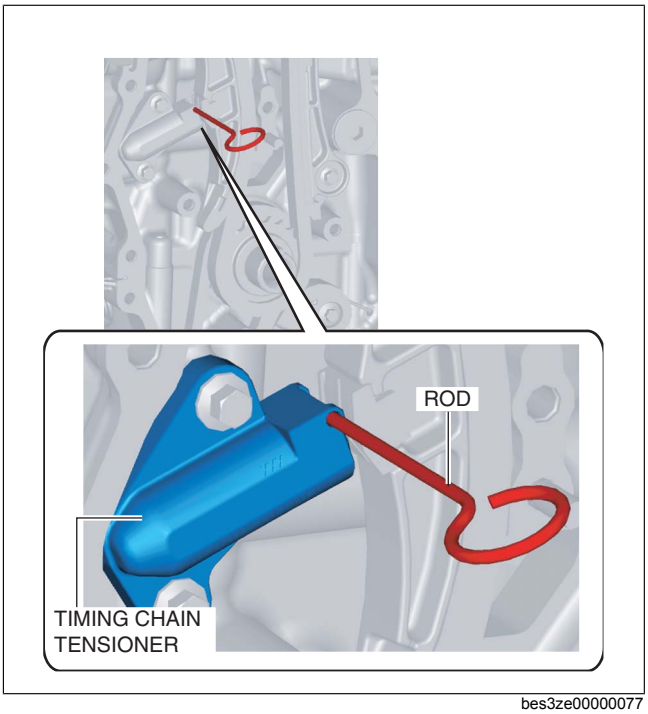
1. Hold the intake camshaft using a wrench on the cast hexagon, and remove the intake camshaft sprocket installation bolt.
2. Remove the intake camshaft sprocket.



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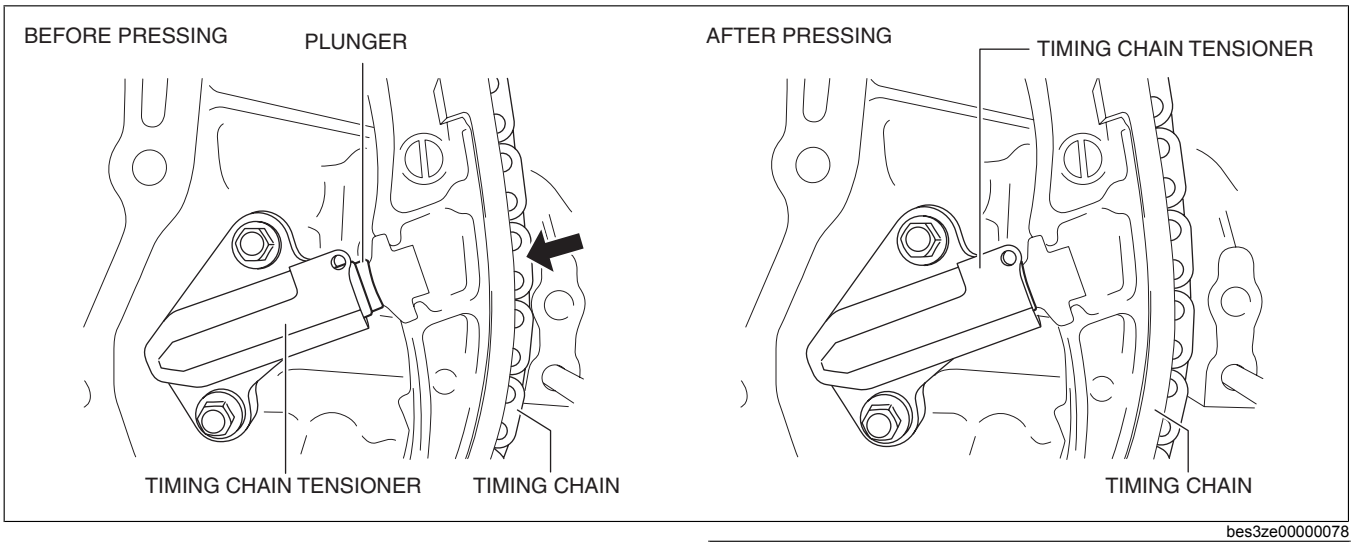
Timing Chain Disassembly Note

1. Loosen the timing chain tensioner using the following procedure:
   (1) Insert a metal rod **(diameter approx. 3.2 mm {0.13 in}, length approx. 60—70 mm {2.4—2.7 in})** into the hole in the body of the timing chain tensioner.



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   (2) Press the timing chain in the direction of the arrow and press on the plunger of the timing chain tensioner.



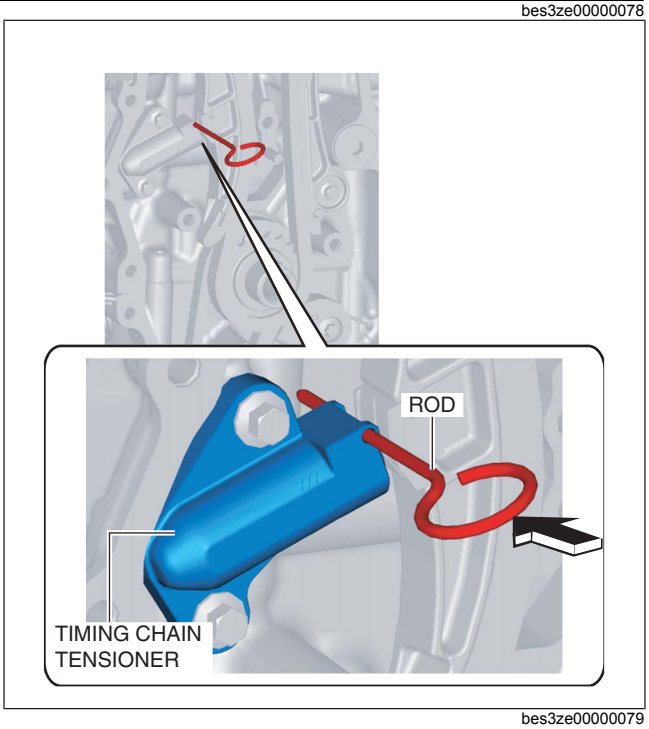
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(3) With the plunger pressed in, further insert the metal rod set in (1) above.

Note
- The rod secures the plunger and releases the tension.

(4) Loosen the power of the hand pressing the plunger and verify that the pressed-in rod does not move.

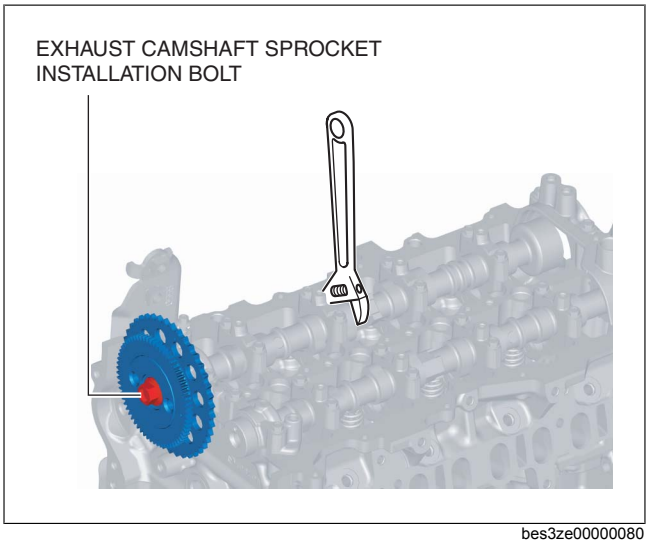
2. Remove the timing chain tensioner and timing chain tensioner arm.
3. Remove the timing chain guide.
4. Remove the timing chain and crankshaft sprocket as a single unit.



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Exhaust Camshaft Sprocket Disassembly Note

1. Hold the exhaust camshaft using a wrench on the cast hexagon, and remove the exhaust camshaft sprocket installation bolt.
2. Remove the exhaust camshaft sprocket.

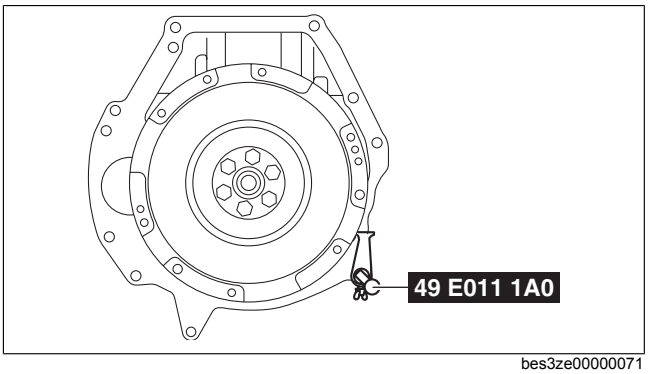


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Oil Pump Driven sprocket Disassembly Note

1. Hold the crankshaft using the **SST**.

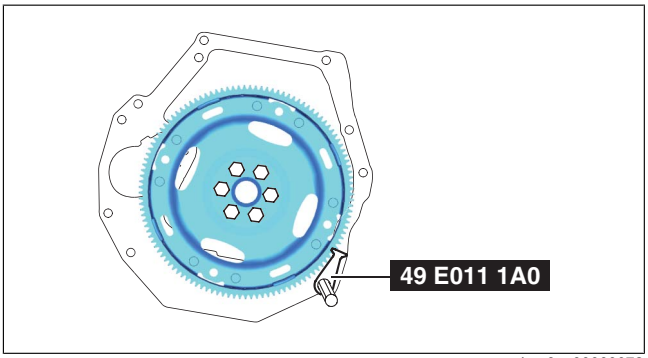
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2. Remove the oil pump driven sprocket.



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